

Backend Engineering Fundamentals - 6 Week Intensive Course

Course Overview

This curriculum will take you from absolute beginner to building production-ready backend applications. By the end of 6 weeks, you'll have a solid foundation in Python backend development, RESTful APIs, databases, authentication, and deployment.

Week 2: Web Fundamentals & Flask Introduction

Learning Objectives

- Understand HTTP protocol, requests, and responses
- Learn about REST architecture and APIs
- Build your first web server with Flask
- Handle routing and request methods

Resources

Videos:

- [HTTP Crash Course](#) by Traversy Media (38 mins)
- [REST API Concepts](#) by WebDevSimplified (8 mins)
- [Flask Tutorial](#) by freeCodeCamp (6 hours - focus on first 3 hours)

Text Resources:

- [Flask Official Documentation - Quickstart](#)
- [Real Python - Flask by Example](#)
- [MDN Web Docs - HTTP Overview](#)

Topics Covered

- HTTP methods (GET, POST, PUT, DELETE)

- Status codes (200, 201, 400, 404, 500)
- Request/Response cycle
- Flask application setup
- Routing and URL parameters
- Request data (JSON, form data, query parameters)
- Response formatting (JSON responses)
- Environment variables and configuration
- Virtual environments (venv)

Week 2 Project: RESTful Book Library API

Build a CRUD API for managing a book collection

Requirements:

- GET /books - List all books
- GET /books/:id - Get a specific book
- POST /books - Add a new book (title, author, ISBN, year)
- PUT /books/:id - Update a book
- DELETE /books/:id - Delete a book
- In-memory storage (Python list/dictionary)
- Proper HTTP status codes
- Input validation
- Error handling

This project tests: Flask routing, HTTP methods, JSON handling, data validation, and REST principles.